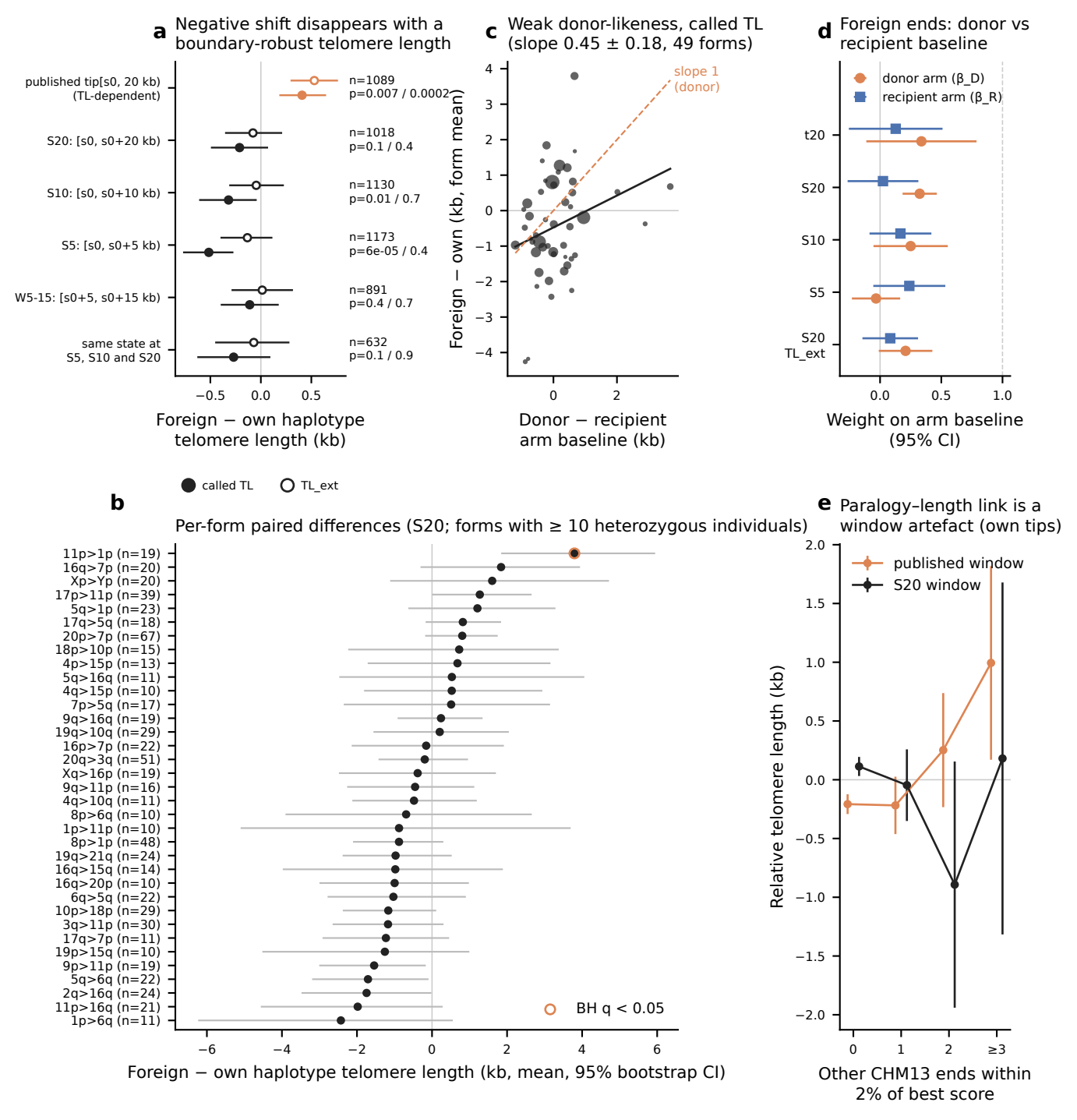
a
Negative shift disappears with a boundary-robust telomere length
published tip[s0, 20 kb) (TL-dependent)
S20: [s0, s0+20 kb)
S10: [s0, s0+10 kb)
S5: [s0, s0+5 kb)
W5-15: [s0+5, s0+15 kb)
same state at S5, S10 and S20
n=1089 p=0.007 / 0.0002
n=1018 p=0.1 / 0.4
n=1130 p=0.01 / 0.7
n=1173 p=6e-05 / 0.4
n=891 p=0.4 / 0.7
n=632 p=0.1 / 0.9
−0.5 0.0 0.5
Foreign − own haplotype telomere length (kb)
c
Weak donor-likeness, called TL (slope 0.45 ± 0.18, 49 forms)
slope 1 (donor)
Foreign − own (kb, form mean)
4 3 2 1 0 −1 −2 −3 −4
0 2
Donor − recipient arm baseline (kb)
d
Foreign ends: donor vs recipient baseline
donor arm (β_D)
recipient arm (β_R)
t20
S20
S10
S5
S20 TL_ext
0.0 0.5 1.0
Weight on arm baseline (95% CI)
b
called TL
TL_ext
Per-form paired differences (S20; forms with ≥ 10 heterozygous individuals)
11p>1p (n=19)
16q>7p (n=20)
Xp>Yp (n=20)
17p>11p (n=39)
5q>1p (n=23)
17q>5q (n=18)
20p>7p (n=67)
18p>10p (n=15)
4p>15p (n=13)
5q>16q (n=11)
4q>15p (n=10)
7p>5q (n=17)
9q>16q (n=19)
19q>10q (n=29)
16p>7p (n=22)
20q>3q (n=51)
Xq>16p (n=19)
9q>11p (n=16)
4q>10q (n=11)
8p>6q (n=10)
1p>11p (n=10)
8p>1p (n=48)
19q>21q (n=24)
16q>15q (n=14)
16q>20p (n=10)
6q>5q (n=22)
10p>18p (n=29)
3q>11p (n=30)
17q>7p (n=11)
19p>15q (n=10)
9p>11p (n=19)
5q>6q (n=22)
2q>16q (n=24)
11p>16q (n=21)
1p>6q (n=11)
BH q < 0.05
−6 −4 −2 0 2 4 6
Foreign − own haplotype telomere length (kb, mean, 95% bootstrap CI)
e
Paralogy–length link is a window artefact (own tips)
published window
S20 window
Relative telomere length (kb)
2.0 1.5 1.0 0.5 0.0 −0.5 −1.0 −1.5 −2.0
0 1 2 ≥3
Other CHM13 ends within 2% of best score